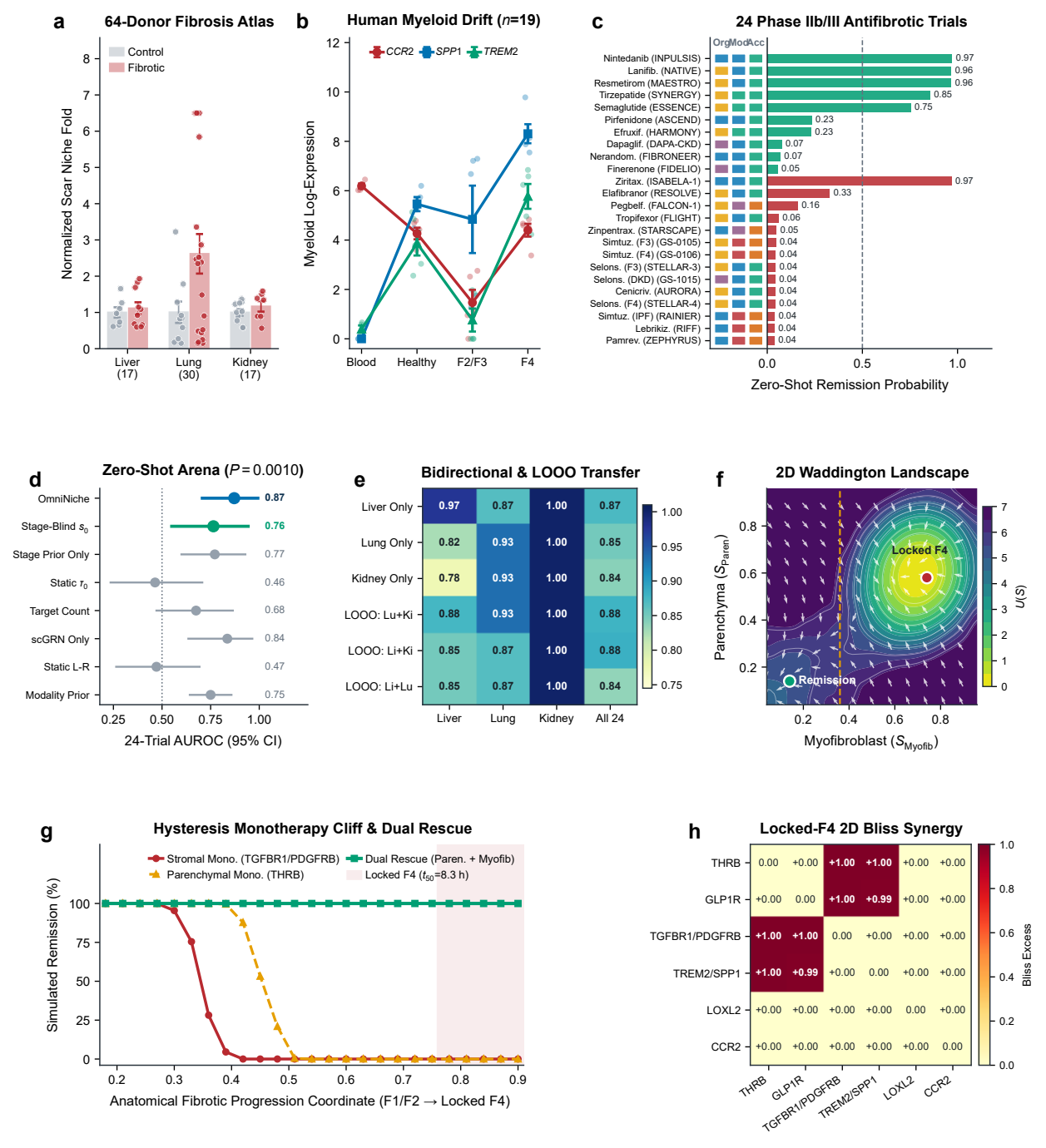

a
64-Donor Fibrosis Atlas
Control
Fibrotic
Normalized Scar Niche Fold
Liver (17)
Lung (30)
Kidney (17)
b
Human Myeloid Drift (n=19)
CCR2
SPP1
TREM2
Myeloid Log-Expression
Blood
Healthy
F2/F3
F4
c
24 Phase IIb/III Antifibrotic Trials
Org
Mod
Acc
Nintedanib (INPULSIS) 0.97
Lanifib. (NATIVE) 0.96
Resmetirom (MAESTRO) 0.96
Tirzepatide (SYNERGY) 0.85
Semaglutide (ESSENCE) 0.75
Pirfenidone (ASCEND) 0.23
Efruxif. (HARMONY) 0.23
Dapaglif. (DAPA-CKD) 0.07
Nerandom. (FIBRONEER) 0.07
Finerenone (FIDELIO) 0.05
Ziritax. (ISABELA-1) 0.97
Elafibranor (RESOLVE) 0.33
Pegbelf. (FALCON-1) 0.16
Tropifexor (FLIGHT) 0.06
Zinpentrax. (STARSCAPE) 0.05
Simtuz. (F3) (GS-0105) 0.04
Simtuz. (F4) (GS-0106) 0.04
Selons. (F3) (STELLAR-3) 0.04
Selons. (DKD) (GS-1015) 0.04
Cenicriv. (AURORA) 0.04
Selons. (F4) (STELLAR-4) 0.04
Simtuz. (IPF) (RAINIER) 0.04
Lebrikiz. (RIFF) 0.04
Pamrev. (ZEPHYRUS) 0.04
Zero-Shot Remission Probability
d
Zero-Shot Arena (P = 0.0010)
OmniNiche 0.87
Stage-Blind s_0 0.76
Stage Prior Only 0.77
Static τ_0 0.46
Target Count 0.68
scGRN Only 0.84
Static L-R 0.47
Modality Prior 0.75
24-Trial AUROC (95% CI)
e
Bidirectional & LOOO Transfer
Liver Only 0.97 0.87 1.00 0.87
Lung Only 0.82 0.93 1.00 0.85
Kidney Only 0.78 0.93 1.00 0.84
LOOO: Lu+Ki 0.88 0.93 1.00 0.88
LOOO: Li+Ki 0.85 0.87 1.00 0.88
LOOO: Li+Lu 0.85 0.87 1.00 0.84
Liver
Lung
Kidney
All 24
f
2D Waddington Landscape
Locked F4
Remission
Parenchyma (S_{Paren})
Myofibroblast (S_{Myofib})
$U(S)$
g
Hysteresis Monotherapy Cliff & Dual Rescue
Stromal Mono. (TGFBR1/PDGFRB)
Parenchymal Mono. (THRB)
Dual Rescue (Paren. + Myofib)
Locked F4 (t_{50}=8.3 h)
Simulated Remission (%)
Anatomical Fibrotic Progression Coordinate (F1/F2 → Locked F4)
h
Locked-F4 2D Bliss Synergy
THRB 0.00 +0.00 +1.00 +1.00 +0.00 +0.00
GLP1R +0.00 0.00 +1.00 +0.99 +0.00 +0.00
TGFBR1/PDGFRB +1.00 +1.00 0.00 +0.00 +0.00 +0.00
TREM2/SPP1 +1.00 +0.99 +0.00 0.00 +0.00 +0.00
LOXL2 +0.00 +0.00 +0.00 +0.00 0.00 +0.00
CCR2 +0.00 +0.00 +0.00 +0.00 +0.00 0.00
THRB
GLP1R
TGFBR1/PDGFRB
TREM2/SPP1
LOXL2
CCR2
Bliss Excess